



Multi-Agent Decentralized Waste Collection and Recycling System

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Problem Context

A dynamic, multi-agent approach offers a scalable solution to the inefficiencies and unpredictability in urban waste collection.

Traditional waste management relies on fixed schedules, leading to:

- Bins overflowing before the scheduled collection.
- Unnecessary truck trips to bins that are still empty.
- Higher operational costs, increased fuel consumption, and traffic congestion.

Cities face additional problems:

- Roadblocks or dynamic obstacles disrupt routes.
- Trucks may run out of fuel, break down, or become unavailable.



Waste types (organic, recyclable) require specialized handling, adding complexity to the logistics.

These challenges demand a smarter, autonomous, adaptive system capable of:

- Real-time decision making
- Efficient route planning
- Coordination between multiple agents (bins, trucks, central)
- Reducing fuel consumption and operational costs

Multi-Agent System Overview

Overall System Architecture

Bin Agents

- Monitor their own waste levels
- Send collection requests when nearing capacity
- Report overflow or abnormal conditions

Truck Agents

- Respond to bin requests and select tasks
- Evaluate route cost, fuel, and load constraints
- Collect waste and return to the central depot for unloading/refueling

Environment

- Stores the map (11×11 grid) and the central's location
- Maintains roadblocks, traffic conditions and NetworkX connectivity
- Holds all agents' states and notifies trucks when the environment changes
- Provides pathfinding data and manages simulation timing

How They Form a Unified System

Bins → Trucks

- Send Call-For-Proposal (CFP) messages when waste $\geq 70\%$ or overflowing.

Trucks → Bins

- Reply with proposals including shortest path, fuel level, and remaining capacity.

Bin → Selected Truck

- Sends accept message to assign the collection task.

Trucks → Central

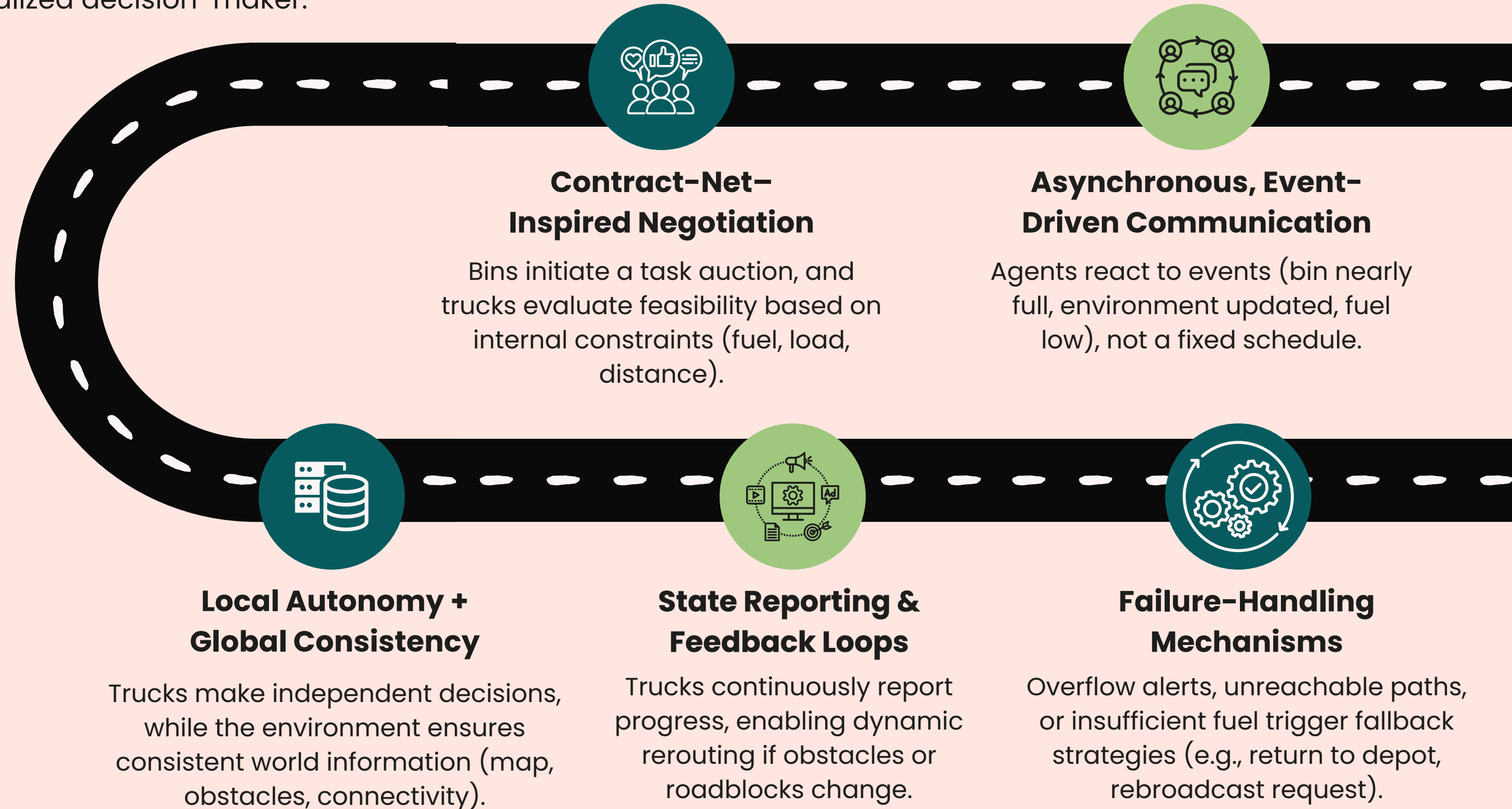
- Return for unloading and refueling when needed.

Environment

- Provides the shared world: map, obstacles, traffic, connectivity, and timing for consistent decision-making.

Interaction and Communication Protocols

Protocols ensure coordination, negotiation, and adaptation so the system behaves coherently without a centralized decision-maker.



Bin Agents — Reactive Architecture

- Simple event-driven logic: detect waste thresholds, trigger CFP.
- No planning; rely entirely on trucks to handle collection decisions.
- Maintain minimal internal state (fill level, overflow status).

Truck Agents – Deliberative Architecture

- Maintain state: position, fuel, load, assigned task, and route.
- Evaluate proposals based on route cost, fuel requirements, and available load capacity.
- Replan routes when roadblocks or environmental changes occur.
- Decide autonomously when to return to the central.

Environment – Centralized World Model (Non-Agent)

- Stores and updates global knowledge: map, obstacles, bin locations, connectivity.
- Provides routing data (NetworkX) and simulation timing.
- Not a decision-maker, but enables consistent decisions among agents.

Collective Behavior

- System-level coordination emerges from local decisions.
- Task assignment follows a distributed auction-like mechanism.
- Agents adapt collectively to dynamic changes (roadblocks, fuel limits, overflow).

Agent Architectures & Decision-Making Strategies

Experimental Metrics Used

To assess the behavior and effectiveness of our multi-agent system, we evaluated performance at three levels: individual trucks, individual bins, and the overall system. The following metrics were collected during five simulations.

1. Truck-Level Metrics (Individual Performance)

- Total waste collected
- Fuel consumed
- Distance traveled
- Number of collaborations (task handovers, assistance)
- Efficiency ratios:
 - Waste per fuel unit
 - Waste per distance unit
- Breakdown / recovery events (if any)

2. Bin-Level Metrics (Local Performance)

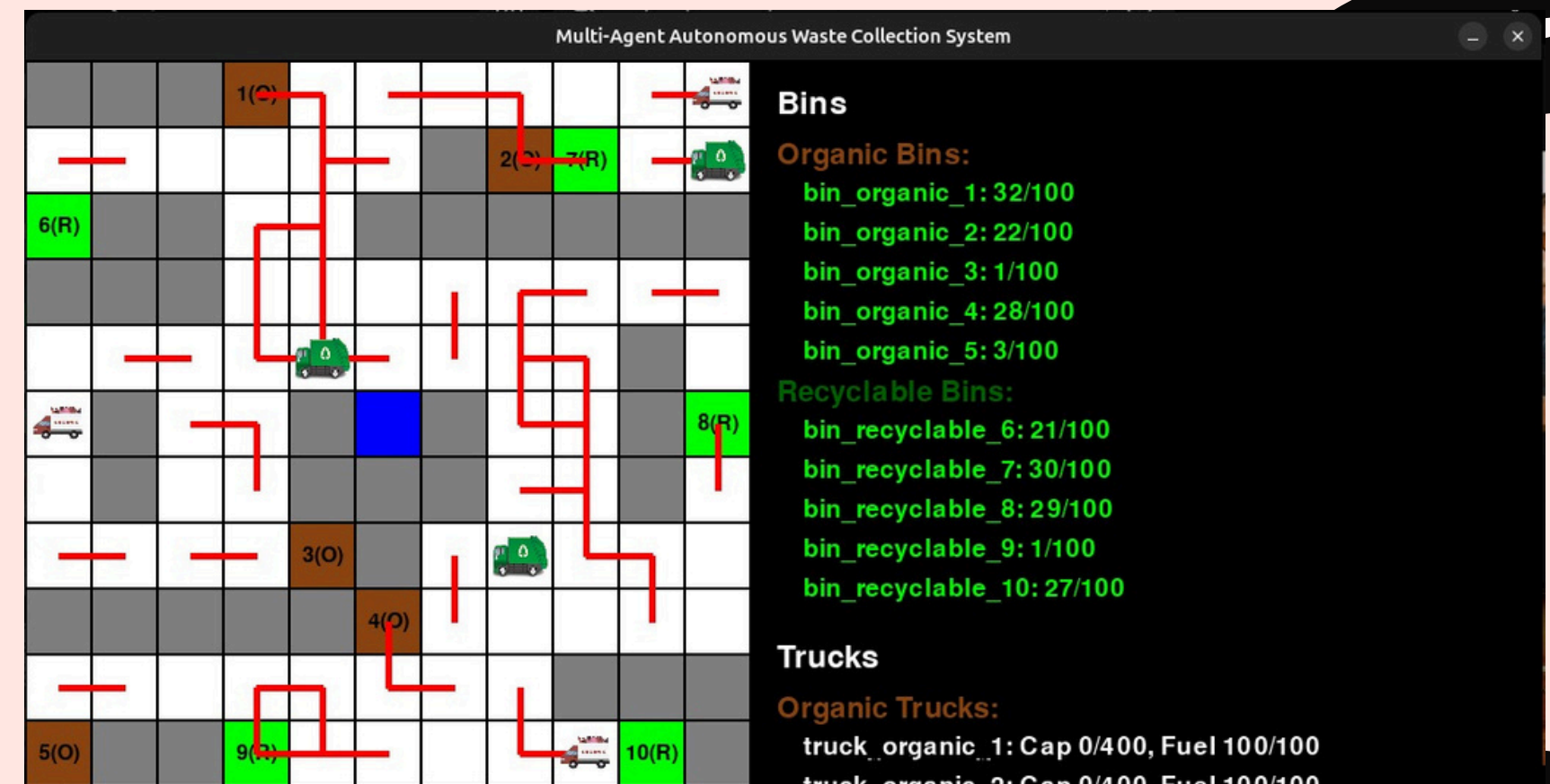
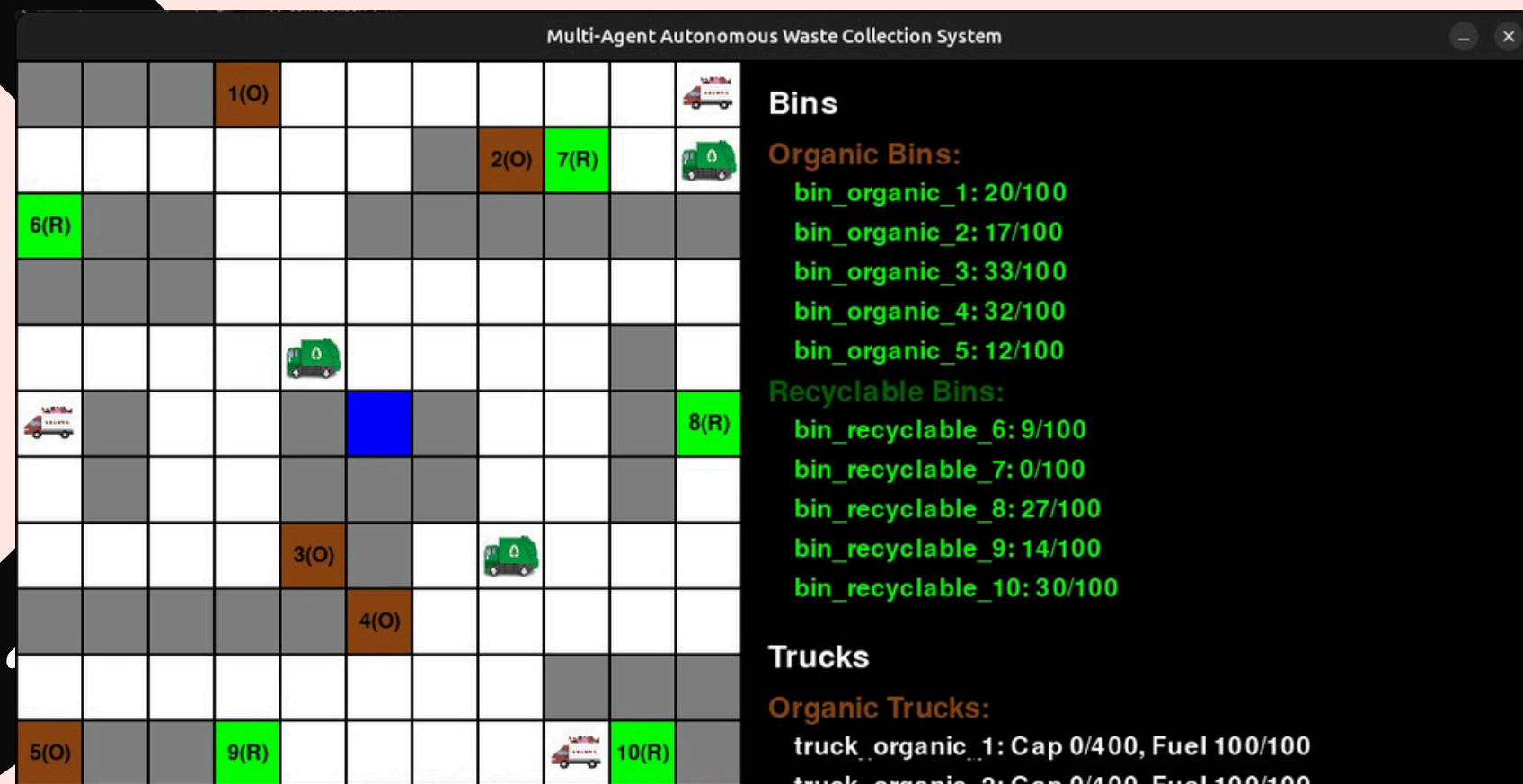
- Total collection events
- CFP-based vs. autonomous collections
- Overflow count
- Collection rate
- CFP response times (min, max, average)
- Time-to-service when near capacity

3. System-Level Metrics (Global Performance)

- Overall collection efficiency
 - % bins collected before overflow
- Total number of collections
 - CFP vs. autonomous
- Average CFP response time
- Total waste collected (organic vs recyclable)
- Recycling rate estimation
- Fuel and distance totals across all trucks
- Collaboration statistics
 - Collaborations per truck
 - Collaborations per breakdown
- Simulation time & stability indicators

Impact of Traffic on System Performance

To assess the impact of environmental complexity, we compared two configurations, with traffic/roadblocks and without traffic. This allows us to measure how dynamic obstacles affect agent efficiency, response times, and overall system performance.



With traffic and roadblocks enabled, we expect reduced efficiency: longer routes, higher fuel use, and slower responses to bin requests. System-wide, this should translate into slower collection cycles and a higher chance of temporary overflows. The comparison highlights how much dynamic obstacles influence agent behavior and overall performance.

Impact of Traffic on System Performance

Analysis of Results

1. Collection Efficiency

- No Traffic: 100% bins collected before overflow (10/10)
- With Traffic: Only 70% collected before overflow (7/10)

Traffic causes delays → more bins reach critical state or overflow.

2. Truck Performance

Fuel Consumption:

- No traffic: 342 total fuel
- With traffic: 843 total fuel
($\approx 2.5\times$ higher)

Distance Traveled:

- No traffic: 342
- With traffic: 542

Trucks travel longer detours and spend more time stuck, consuming more fuel.

3. Waste Collection

- No traffic: 2328 units collected
- With traffic: 2348 units collected (slightly higher)

Despite delays, trucks still collect a similar total amount—but less efficiently.

4. CFP (Call-For-Proposal) Activity

- No traffic: 0 CFP collections (all autonomous)
- With traffic: 11 CFP collections

Traffic causes unpredictable delays → bins request help more often.

5. Bin Behavior

Overflows:

- No traffic: 0
- With traffic: 3 overflows

CFP Response Times:

- No traffic: N/A
- With traffic: Average ~21s, worst case 64s

Travel times become inconsistent → slower service, especially for remote bins.

6. System Efficiency Metrics

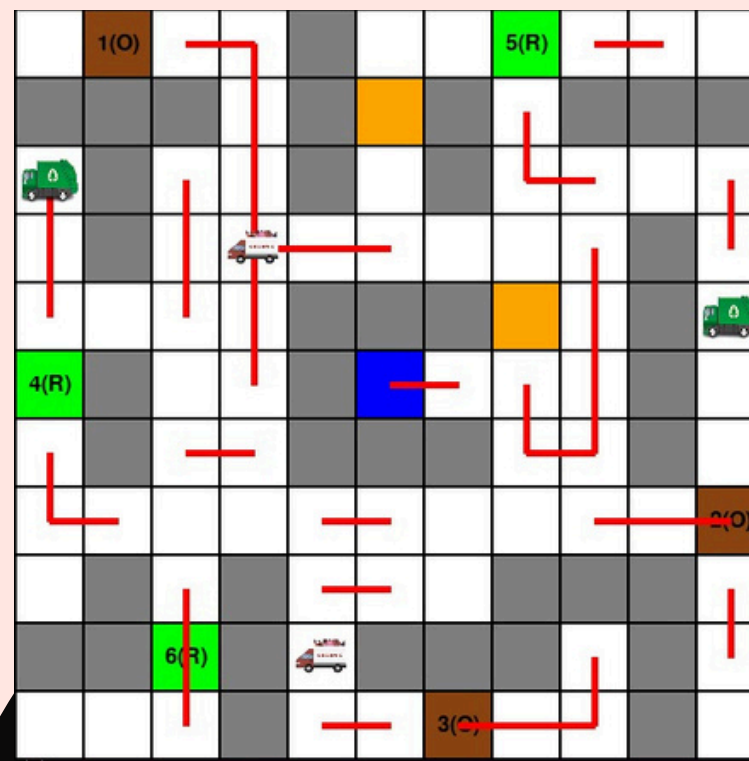
Waste / Fuel Efficiency:

- No traffic: 6.81 waste per fuel
 - With traffic: 2.79 waste per fuel
- ~60% reduction in efficiency

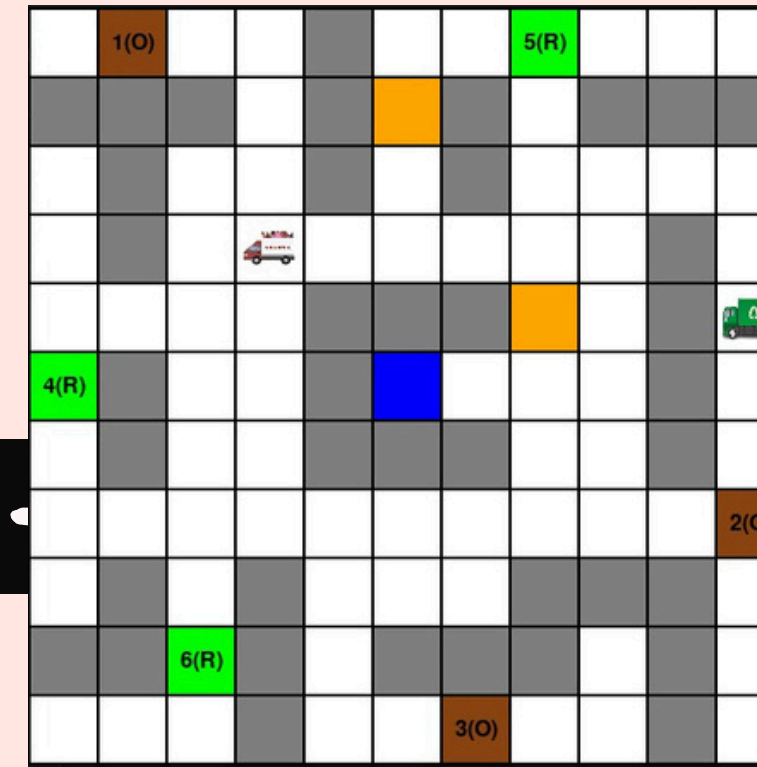
Waste / Distance Efficiency:

- Decreased from 6.81 → 4.33
- Efficiency drops sharply due to detours and congestion.

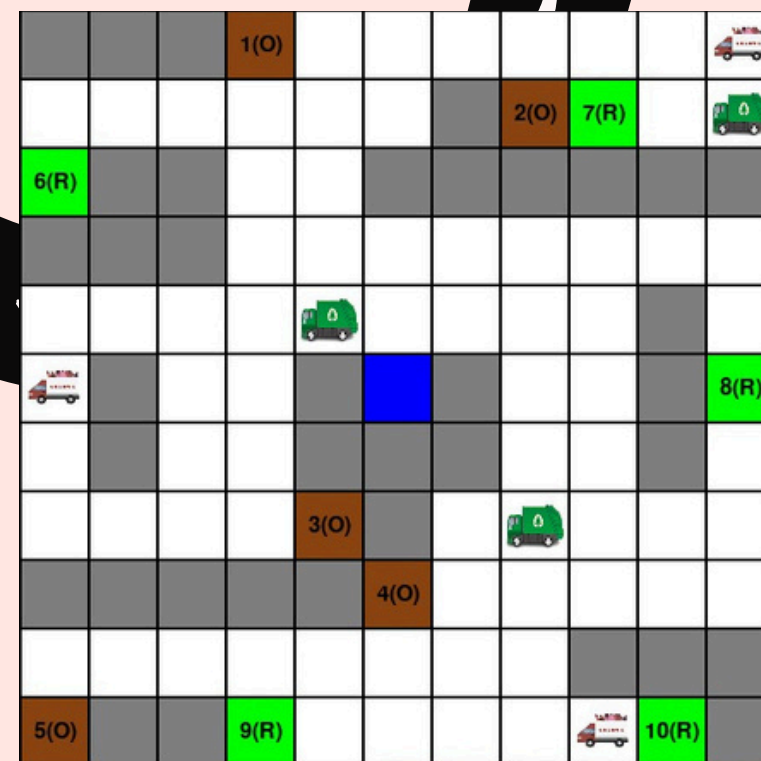
Experimental Configurations



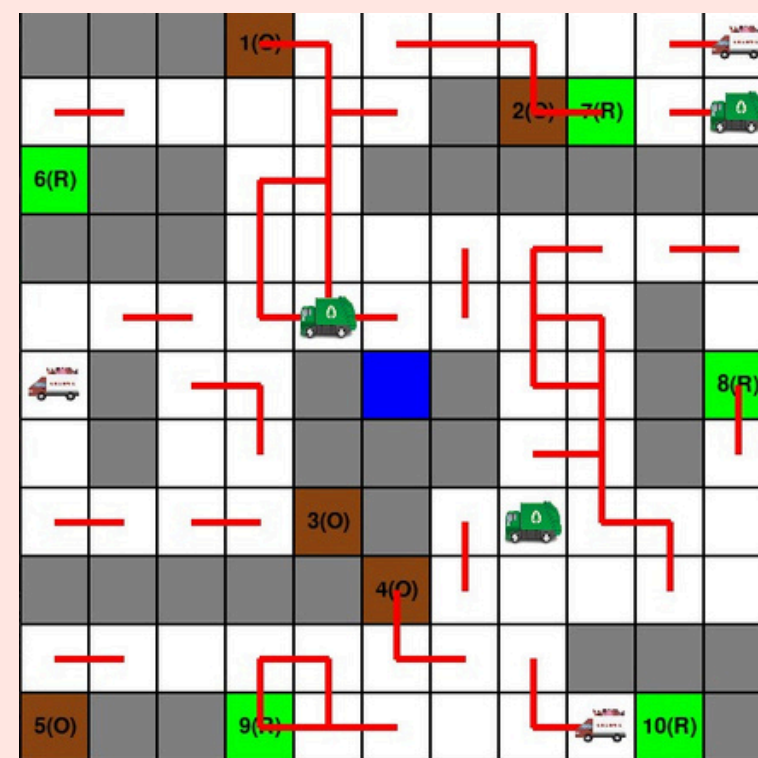
Configuration 1



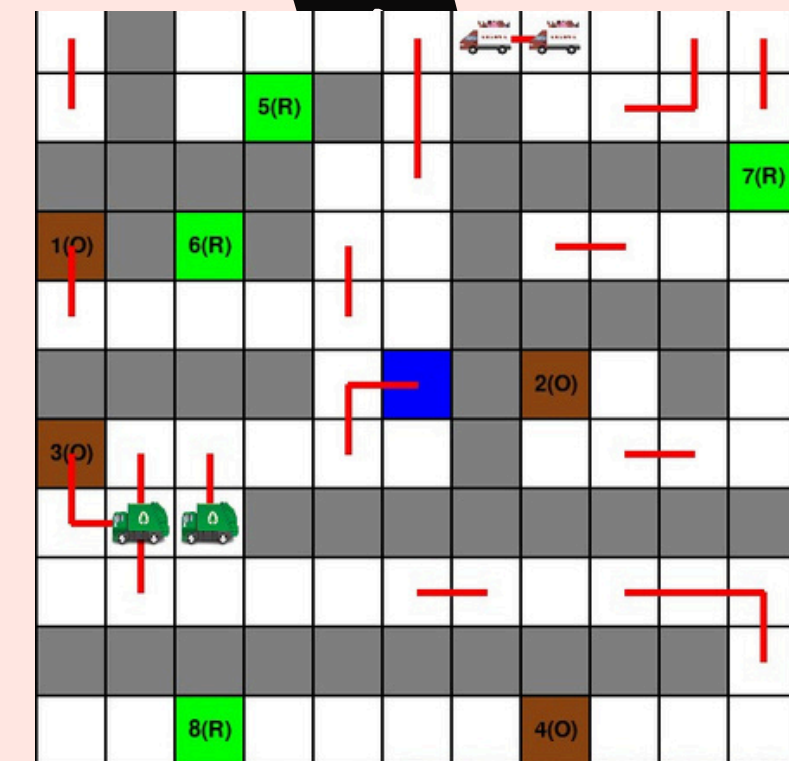
Configuration 2



Configuration 3



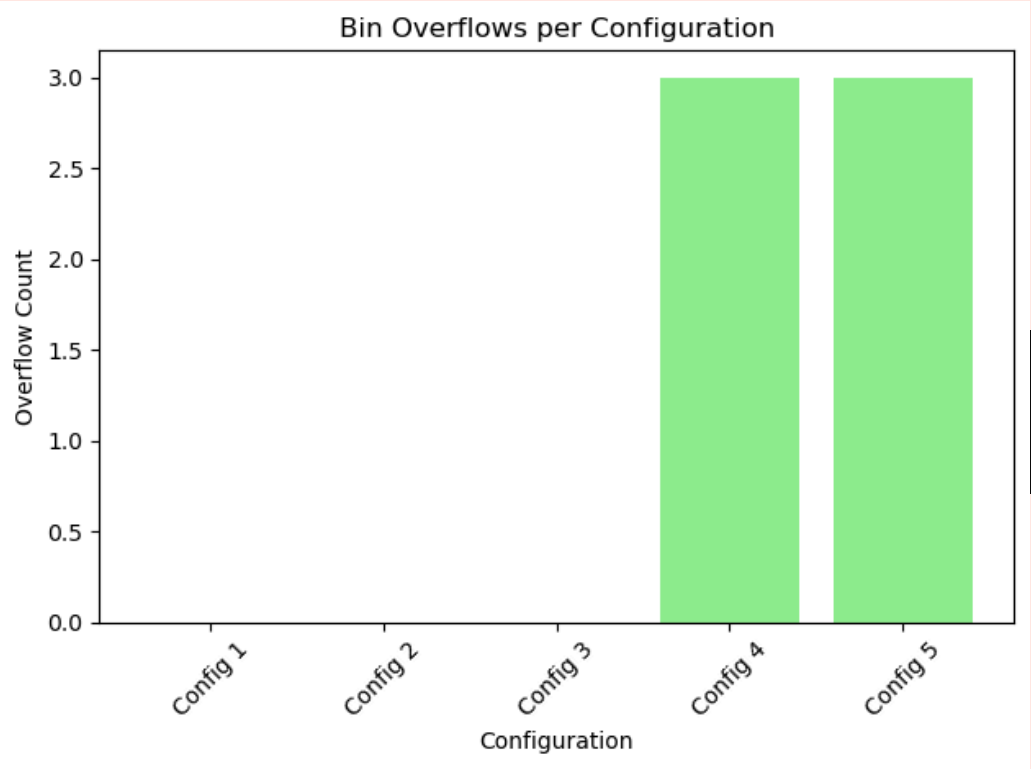
Configuration 4



Configuration 5

Analysis of Results

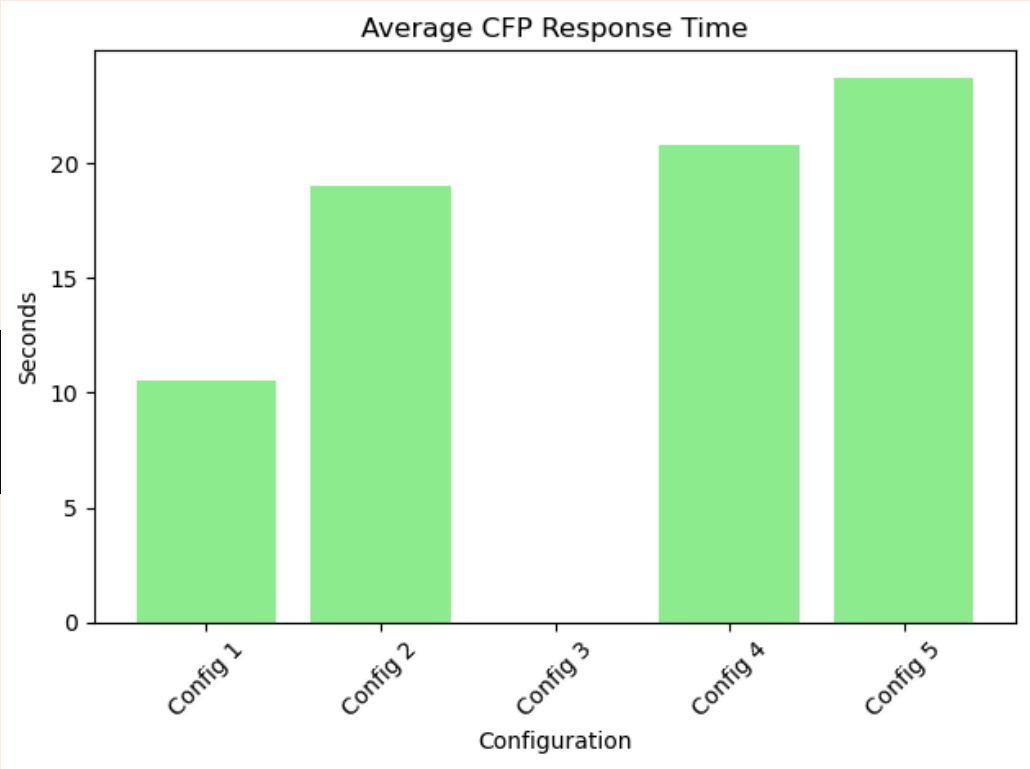
Comparison between different configurations



Bin Overflow per Configuration

Configurations 1–3 had zero overflows, showing sufficient truck capacity and timely collections.

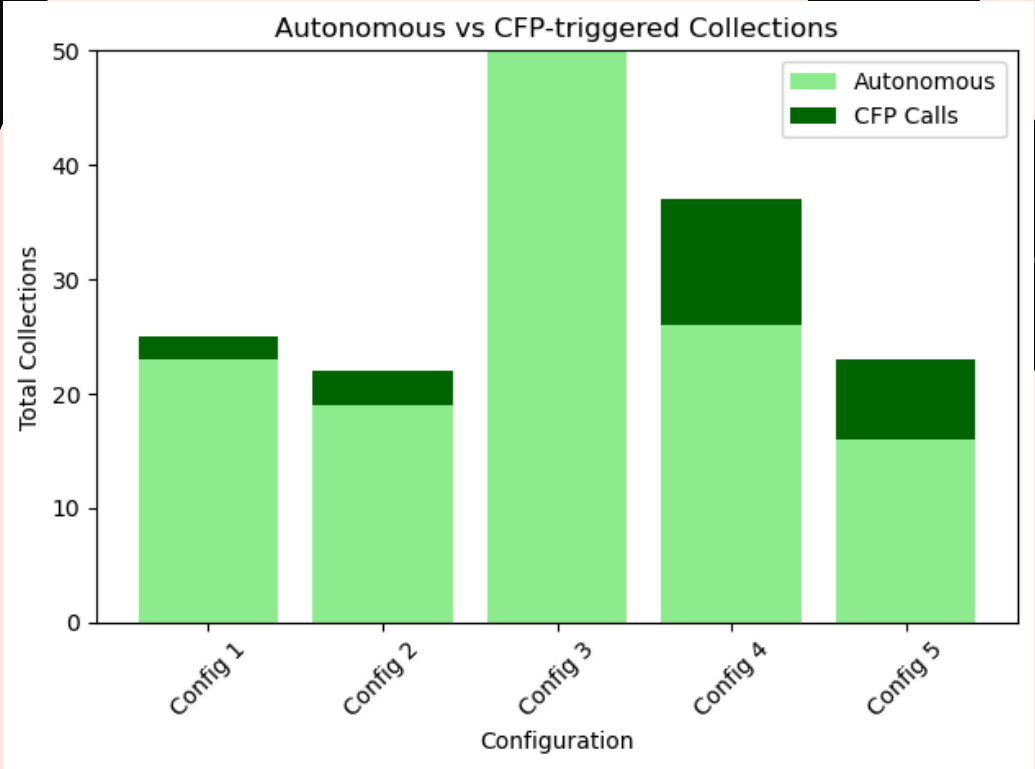
Configurations 4 and 5 each had three overflows, indicating reduced responsiveness, likely due to fewer trucks, altered routes, or longer distances.



Average CFP Response Time

Configuration 1 responded fastest (~10.5s), while Configuration 5 was slowest (~23.5s).

Response times increase from configurations 1 to 5, suggesting higher workload or limited truck availability in later setups.



Autonomous vs CFP-triggered Collections

Configuration 3 had the most autonomous collections (50) and few CFP-triggered ones, showing efficient proactive routing.

Configurations 4 and 5 relied more on CFP-triggered collections, reflecting reactive behaviour.

Configurations 1 and 2 were intermediate, with balanced autonomous and CFP collections.

Conclusions and Critical Analysis

The multi-agent system successfully coordinates bins, trucks, and the environment through decentralized decision-making and dynamic negotiation.

Event-driven and contract-net-inspired protocols enable adaptive responses to real-time changes (fuel, roadblocks, overflow).

In stable environments (no traffic), the system achieves 100% timely collection, 0 overflows, and high fuel efficiency.

Critical Analysis

- With traffic and roadblocks, routes became longer and fuel usage increased, as expected in more complex environments.
- A few delays and overflows occurred, suggesting room for better anticipation of dynamic obstacles.
- CFP requests rose in tougher scenarios, showing the system's reactive support mechanisms are working as intended.



Future Work

- Add predictive route planning to better handle traffic and congestion.
- Improve truck coordination to balance workload more effectively.
- Use learning or heuristic methods to help agents anticipate obstacles.
- Test the system on larger maps and more agents to evaluate scalability.
- Include different truck types or energy models for greater realism.